

Database Project

Smart Hostel Management System



Submitted To:

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1 Introduction

The Smart Hostel Management System (SHMS) is designed to automate and secure daily hostel operations including student attendance management, room allocation, complaint handling, exit/entry management, and warden oversight. The system ensures that only students inside a predefined geofence boundary can mark attendance or request exit during restricted hours. Warden manages all allocations, approvals, and complaint resolutions. Security personnel views approved exits via a dedicated screen. Face selfies are taken at attendance and exit time but stored only temporarily for random verification by the warden they are not compared with any reference photo. The system also maintains a complete audit trail of all changes made by the system administrator. This report presents the complete Entity-Relationship Diagram in Crow's Foot notation, along with entities, attributes, business rules, and cardinalities as required for Milestone 2.

2 Problem Statement

Traditional hostel management systems face multiple operational problems because most activities are performed manually. Major problems include:

- Proxy attendance and unauthorized exit from hostel
- Lack of real-time tracking of student movement
- No audit trail for room changes or exit approvals
- Delayed communication of absentee lists to wardens
- Difficulty verifying if a student is inside hostel premises during attendance or exit requests
- Manual record keeping leading to data redundancy and errors

The proposed system solves these by enforcing geofence-based verification, temporary selfie capture, attendance marking (Present/Absent/OnLeave), and role-based access for warden, security, and admin.

3 Objectives

3.1 General Objective

To develop a secure, semi-automated hostel management system that uses geofence technology and role-based workflows to improve attendance integrity, exit control, and administrative oversight.

3.2 Specific Objectives

- Allow students to mark daily attendance only when inside a predefined geofence boundary

- Enable students to submit two types of exit requests: short exit (needs approval during restricted hours) and leave (auto-marks attendance as OnLeave until return)
- Provide a view for security personnel showing only approved short-exit applications
- Automatically detect student return via geofence entry and update exit logs
- Automatically generate absent and on-leave lists for the warden after the 10:00 PM attendance deadline
- Maintain full audit of all actions performed by the system administrator
- Store selfies temporarily for random warden verification

4 System Overview

The Smart Hostel Management System contains four major user types:

- **System Administrator:** Manages system-level operations, user accounts (Student, Warden, Security), geofence boundaries, notifications, and device registrations. All changes are logged in SystemChangeLog.
- **Warden:** Manages hostel operations including attendance monitoring, exit application approval, complaint resolution, room allocation, and report generation.
- **Security Personnel:** Views approved exit applications on a gate screen and allows exit only for approved students during restricted hours.
- **Student:** Marks attendance (with geofence + selfie), submits exit applications (short exit or leave), submits complaints, and views notifications.

The system uses geofence detection for return confirmation and sends automatic alerts for late returns to both student and warden.

5 Entities Identification

The following entities were identified after complete system analysis:

5.1 Supertype

- **User** (supertype with common attributes)

5.2 Subtypes (inheriting from User)

- Student
- Warden
- SecurityPersonnel

- SystemAdministrator

5.3 Main Entities

- Room
- RoomAllocation
- AttendanceRecord
- ExitApplication
- EntryExitLog
- GeofenceBoundary
- DeviceRegistration
- Notification
- Complaint
- SystemChangeLog

Total: 15 entities (1 supertype + 4 subtypes + 10 main entities)

6 Supertype and Subtype Analysis

The system uses a supertype-subtype structure where **User** is the supertype containing common attributes (UserID, Name, Email, PasswordHash, Phone, Role, CreatedAt). Four subtypes inherit from User:

- **Student**: Adds RollNumber, Department, Semester, ProfilePhoto
- **Warden**: Adds OfficePhone, HireDate
- **SecurityPersonnel**: Only SecurityID (single PC, single gate)
- **SystemAdministrator**: Only AdminID (Admin have full access)

This design follows disjoint specialization (a User is exactly one subtype) and partial specialization (not all User instances are in subtypes).

7 Entity Attributes

7.1 User (Supertype)

Attribute	Description
UserID (PK)	Unique identifier for all users
Name	Full name of the user
Email	Email address (unique)
Password	Encrypted password
Phone	Contact number
Role	User role (Student/Warden/Security/Admin)
CreatedAt	Account creation timestamp

7.2 Student (Subtype of User)

StudentID (PK)	Unique identifier for each Student
RollNumber	Student's roll number
Department	Academic department
Semester	Current semester
ProfilePhotoPath	Profile picture (display only)

7.3 Warden (Subtype of User)

WardenID (PK)	Unique identifier for each warden
OfficePhone	Warden's office contact
HireDate	Date of joining

7.4 SecurityPersonnel (Subtype of User)

SecurityID (PK)	Unique identifier for SecurityPersonnel
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Note: Single SecurityPersonnel, single gate — no extra attributes needed.

7.5 SystemAdministrator (Subtype of User)

AdminID (PK)	Unique identifier for each Admin
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Note: All admins have full system access.

7.6 Room

RoomID (PK)	Unique room identifier
RoomNumber	Room number
Capacity	Maximum students allowed
CurrentOccupancy	Current number of students
RoomStatus	Availability status

7.7 RoomAllocation

Attribute	Description
AllocationID (PK)	Unique identifier for each room allocation
StudentID (FK)	Identifier for each Student
RoomID (FK)	Identifier for each Room
AllocationType	Type of allocation (e.g., 'Manual', 'ByWarden')
AllocationDate	Date and time when allocation was made
Status	Current status (e.g., 'Active', 'Ended')

7.8 AttendanceRecord

AttendanceID (PK)	Unique attendance identifier
StudentID (FK)	References Student.StudentID
Date	Attendance date
VerificationTime	Time of marking (NULL if auto-marked)
Status	Present / Absent / OnLeave
VerificationMethod	Selfie+Geofence / Auto-Leave / Auto-Absent
GPSLocation	Location where attendance marked
SuspiciousFlag	TRUE if anomaly detected

7.9 ExitApplication

Attribute	Description
ApplicationID (PK)	Unique identifier
StudentID (FK)	References Student.StudentID
Destination	Location where student is going
IsOnLeave	TRUE = leave, FALSE = short exit
Reason	Reason for exit
DepartureTime	Planned departure time
ExpectedReturnTime	Expected return (NULL for leave)
RestrictedTimeFlag	TRUE if restricted hours (needs approval)
SubmittedAt	Submission timestamp
ApprovalStatus	Pending / Approved / Rejected / Active
WardenID (FK)	References Warden.WardenID
WardenRemarks	Warden's comments

7.10 EntryExitLog

Attribute	Description
LogID (PK)	Unique identifier
ApplicationID (FK)	References ExitApplication.ApplicationID (NULL for free-hour exits)
StudentID (FK)	References Student.StudentID
ExitTime	Actual exit timestamp
ActualReturnTime	Actual return timestamp (set by geofence)
Selfie	Temporary selfie path (24-hour retention)
LateReturnAlert	TRUE if return late, FALSE otherwise

7.11 GeofenceBoundary

Attribute	Description
GeofenceID (PK)	Unique identifier for each geofence boundary
BoundaryName	Name of the geofence area (e.g., "Hostel Premises")
LattudePolygon	Coordinates defining the geofence area
RadiusMeters	Radius of the geofence in meters

7.12 DeviceRegistration

Attribute	Description
DeviceID (PK)	Unique identifier for each registered device
StudentID (FK)	student who owns the device
DeviceToken	Unique device token for identification
DeviceType	Type of device (iOS, Android, Web)
LastActiveAt	Timestamp of last activity
RegisteredAt	Timestamp of device registration

7.13 Notification

Attribute	Description
NotificationID (PK)	Unique identifier
UserID (FK)	recipient of the notification
Type	Notification category
Message	Notification content
SentAt	Sent timestamp
IsRead	Read status (TRUE/FALSE)

7.14 Complaint

Attribute	Description
ComplaintID (PK)	Unique identifier
StudentID (FK)	References Student.StudentID
Title	Complaint subject
Description	Detailed complaint description
ComplaintType	Category of complaint
Attachment	Attached file (optional)
Status	Pending / In Progress / Resolved / Rejected
SubmittedAt	Submission timestamp
ResolvedAt	Resolution timestamp
WardenRemarks	Warden's comments or resolution notes

7.15 SystemChangeLog

Attribute	Description
LogID (PK)	Unique identifier
AdminID (FK)	References SystemAdministrator.AdminID
Action	ACTIVE/UPDATE/DELETE/SUSPEND
EntityName	Entity that was changed
RecordID	ID of the affected record
Time/Date	Timestamp of the action
Description	Brief description of the change (optional)

8 Relationships and Cardinalities

The following table shows all relationships in the Smart Hostel Management System with their names and cardinalities in **Chen notation** (min, max).

From Entity	To Entity	Relationship Name	Cardinality
User	Student	is a	(1,1) : (0,1)
User	Warden	is a	(1,1) : (0,1)
User	SecurityPersonnel	is a	(1,1) : (0,1)
User	SystemAdministrator	is a	(1,1) : (0,1)
Student	AttendanceRecord	marks	(1,1) : (0,N)
Student	ExitApplication	submits	(1,1) : (0,N)
Student	EntryExitLog	performs	(1,1) : (0,N)
Student	Complaint	submits	(1,1) : (0,N)
Student	DeviceRegistration	registers	(1,1) : (0,N)
Student	RoomAllocation	has	(1,1) : (0,N)
Room	RoomAllocation	assigned to	(1,1) : (0,N)
Warden	ExitApplication	approves	(1,1) : (0,N)
Warden	Complaint	resolves	(1,1) : (0,N)
Warden	RoomAllocation	performs	(1,1) : (0,N)
Warden	Room	manages	(1,1) : (0,N)
SecurityPersonnel	ExitApplication	views (approved only)	(1,1) : (0,N)
SystemAdministrator	SystemChangeLog	creates	(1,1) : (0,N)
SystemAdministrator	GeofenceBoundary	configures	(1,1) : (0,N)
SystemChangeLog	Student	logs changes to	(0,N) : (1,1)
SystemChangeLog	Warden	logs changes to	(0,N) : (1,1)
SystemChangeLog	SecurityPersonnel	logs changes to	(0,N) : (1,1)
SystemChangeLog	GeofenceBoundary	logs changes to	(0,N) : (1,1)
SystemChangeLog	Notification	logs changes to	(0,N) : (1,1)
SystemChangeLog	DeviceRegistration	logs changes to	(0,N) : (1,1)
ExitApplication	EntryExitLog	generates	(0,1) : (0,1)
GeofenceBoundary	EntryExitLog	triggers return	(1,1) : (0,N)
GeofenceBoundary	AttendanceRecord	verifies location	(1,1) : (0,N)
DeviceRegistration	AttendanceRecord	used to mark	(1,1) : (0,N)
DeviceRegistration	EntryExitLog	tracks	(1,1) : (0,N)
EntryExitLog	Notification	triggers alert	(1,1) : (0,N)
Notification	User	sent to	(0,N) : (1,1)

9 Business Rules

1. **BR-1:** Attendance can be marked only if the student's device GPS is inside the active GeofenceBoundary.

2. **BR-2:** A fresh selfie is captured at attendance, exit, and return time but stored temporarily (for random warden verification).
3. **BR-3:** Daily attendance deadline is 10:00 PM. After that, the system automatically marks missing students as Absent or OnLeave (active leave application).
4. **BR-4:** An ExitApplication with IsOnLeave = TRUE automatically marks the student OnLeave until geofence return is detected.
5. **BR-5:** Exit applications during restricted hours require warden approval; during free hours they are auto-approved.
6. **BR-6:** Only approved exit applications appear on the security guard's view.
7. **BR-7:** Geofence entry automatically updates ActualReturnTime in EntryExitLog and ends an active leave.
8. **BR-8:** If ActualReturnTime exceeds ReturnTime, a LateReturnAlert is triggered and two Notifications are sent (to student and warden).
9. **BR-9:** Every admin action (create/update/delete on Student, Warden, SecurityPersonnel, GeofenceBoundary, Notification, DeviceRegistration) is recorded in SystemChangeLog.
10. **BR-10:** Only one warden exists in the system; that warden performs all allocations, approvals, and complaint resolutions.
11. **BR-11:** Security personnel only view approved students applications but do not modify them and allow approved students to exit gate.
12. **BR-12:** A student can have multiple registered devices, and any of devices can be used.
13. **BR-13:** A student can have only one active RoomAllocation at a time.
14. **BR-14:** When a room allocation ends (student moves out), IsCurrent is set to FALSE and DeallocatedAt is recorded.
15. **BR-15:** All notifications are linked to User supertype, not directly to subtypes.

10 ERD Design and Representation

The ERD is drawn using **Crow's Foot notation** as required by the milestone:

- Entities are represented by rectangles
- Attributes are listed inside each entity (primary keys underlined)
- Relationships are drawn as lines with cardinality symbols
- The supertype User is connected to its four subtypes with a disjoint, total specialization symbol (circle with double line)

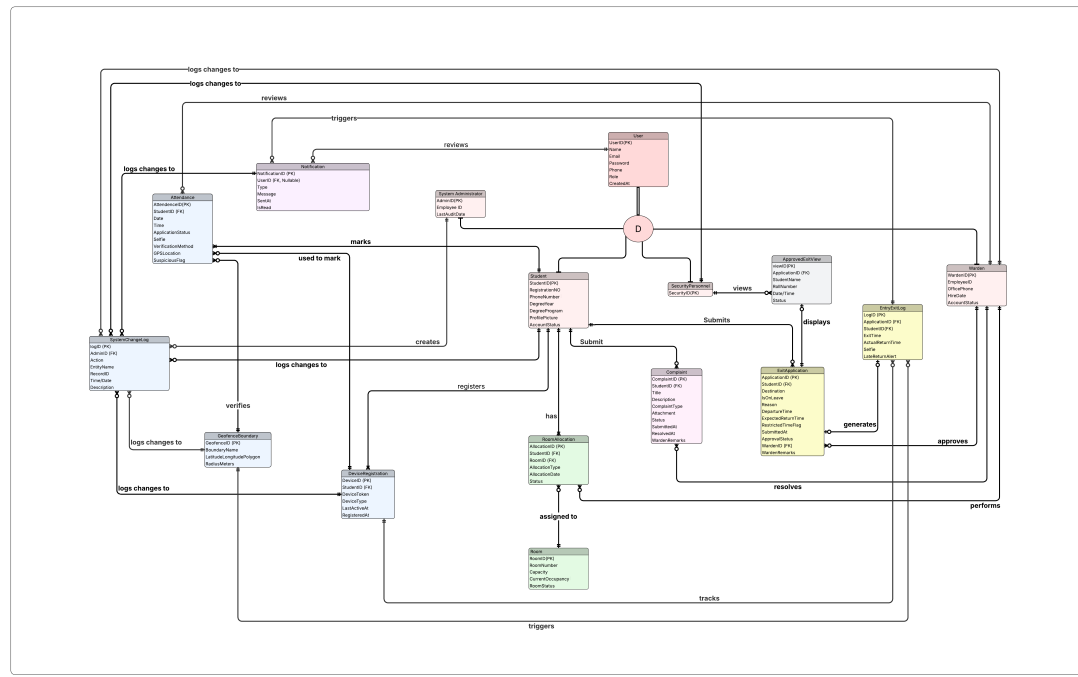


Figure 1: Entity Relationship Diagram of SHMS

11 Advantages of the Proposed System

- Reduces paperwork through digital records
- Improves hostel security with geofence-based verification
- Provides centralized data management with full audit trail
- Improves communication efficiency via automated notifications
- Enables quick attendance and exit report generation
- Reduces human errors in attendance marking
- Provides efficient complaint handling with status tracking
- Enables fast application approval process for restricted-time exits
- Maintains organized room allocation
- Prevents proxy attendance through temporary selfie capture

12 Conclusion

The Smart Hostel Management System provides an efficient, secure, and centralized solution for hostel management operations. The system automates attendance management (with geofence + selfie), room allocation with full history tracking, complaint handling, entry/exit monitoring, and application approval management. The proposed database

design uses a supertype-subtype structure (User with four subtypes) and includes 15 fully normalized entities. The SystemChangeLog entity ensures complete audit trail for all admin actions. The Entity Relationship Diagram developed in this milestone provides a strong foundation for implementation of the Smart Hostel Management System.

13 Teamwork & Professional Conduct

- Regular coordination meetings were held
- Each member actively contributed and reviewed the final deliverables
- All sources (draw.io, Lucidchart) and AI assistance are disclosed in the AI appendix

14 AI Prompts Appendix

The following AI prompts were used during project analysis and report preparation:

1. Identify entities and attributes for hostel management system with geofence and temporary selfie
2. Generate business rules for attendance deadline, leave handling, and late return alerts
3. Create Crow's Foot ERD relationships with proper cardinalities

All technical decisions (entities, relationships, cardinalities, business rules) were made by the team after analysis. AI was used only for help or clearing the confusions, LaTeX formatting, and validation of design consistency.

GitHub Repository

The complete project source code, database schema, and ERD files are available on GitHub at the following link:

<https://github.com/Muhammad-Ahmad-99/Smart-Hostel-Management-System.git>

The repository contains:

- Complete SQL database schema
- ERD diagrams (JPEG format)
- Project documentation